

E10 · Federated-learning methods as the Phase-1 θ aggregator

A · distance to the pooled estimator after 10 rounds $\ w_{10} - w^*\ /\ w^*\ $ (tuned; offline, N=8, 100 rounds of data)								
Sufficient statistics (exact)	0	0	0	0	0	0	0	0
Split learning	0	0	0	0	0	0	0	0
One-shot FedAvg	1e-09	1e-09	1e-09	1e-09	4e-02	6e-02	1e-01	6e-02
Fed. distillation	1e-09	1e-09	1e-09	1e-09	4e-02	6e-02	1e-01	6e-02
FedSGD / gradient-based	0	0	0	0	1e-06	1e-05	1e-06	1e-06
FedAvg	0	0	0	0	7e-04	7e-03	7e-04	7e-04
FedAvg, steps $\propto$ data	0	0	0	0	7e-04	3e-02	7e-04	7e-04
FedProx	0	0	0	0	2e-02	4e-02	5e-02	3e-02
FedNova	0	9e-02	0	0	1e-02	2e-02	4e-02	2e-02
FedAvgM	2e-02	2e-02	2e-02	2e-02	2e-02	2e-02	2e-02	2e-02
FedAdam	3e-01	3e-01	3e-01	3e-01	3e-01	3e-01	3e-01	3e-01
FedYogi	3e-01	3e-01	3e-01	3e-01	3e-01	3e-01	3e-01	3e-01
SCAFFOLD	0	0	0	0	1e-04	2e-03	1e-04	1e-04
FedDyn	0	2e-03	0	0	5e-03	3e-02	5e-03	5e-03
stein-iid stein-participation stein-covariate stein-concept ls-iid ls-participation ls-covariate ls-concept								

B · bandit: network regret $\div$ exact pooling (blue = better) Phase 2 fixed at exact sync · N=8, T=10,000, 8 trials								
Sufficient statistics (exact)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Split learning	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
One-shot FedAvg	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Fed. distillation	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FedSGD / gradient-based	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FedAvg	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FedAvg, steps $\propto$ data	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FedProx	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FedNova	1.00	1.00	1.12	1.10	1.00	1.00	1.00	1.00
FedAvgM	1.00	1.00	1.00	1.00	1.34	1.00	1.00	1.00
FedAdam	0.93	0.63	1.08	0.97	1.41	0.79	0.93	0.94
FedYogi	0.96	0.62	0.74	1.00	1.20	0.80	0.90	0.93
SCAFFOLD	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FedDyn	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Sufficient statistics (exact) [ls]	1.08	0.27	0.67	0.43	0.95	0.89	0.91	0.94
FedAvg [ls]	1.05	0.29	0.64	0.44	0.97	1.10	0.91	0.89
SCAFFOLD [ls]	1.01	0.27	0.65	0.46	0.95	1.02	0.91	0.90
FedDyn [ls]	1.05	0.29	0.64	0.58	0.97	1.10	0.95	0.89
N x ZoomSIB-UCB (no comm.)	3.35	2.01	2.67	3.50	2.95	3.14	0.79	0.65
iid-quad iid-zigzag participation-quad participation-zigzag covariate-quad covariate-zigzag concept-quad concept-zigzag								

